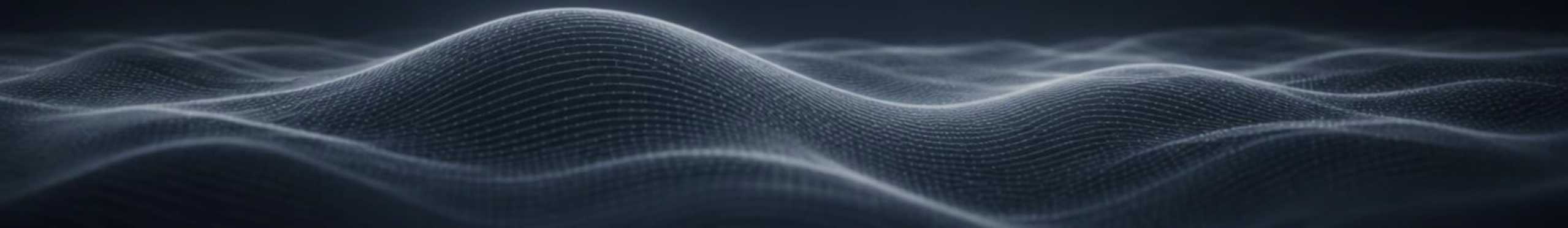
RETAIL ANALYTICS CASE STUDY | END-TO-END PIPELINE

Customer Shopping Behavior Analysis

An end-to-end analytics project connecting transaction data, customer behavior, and commercial decisions.

Python ETL · PostgreSQL Data Modeling · Power BI Interactive Reporting
Executive & Analytics Stakeholder Review



EXECUTIVE SUMMARY

3,900 customers reveal a broad but under-converted opportunity

Average order

$59.76

Average catalog rating

3.75 / 5

Subscription rate

27%

1,053 users

Commercial signal

73%

of buyers remain outside the subscription program

PROJECT LIFECYCLE

The project turns raw transactions into decisions

STEP 01

Clean in Python

Audit raw attributes, profile schemas, and standardize messy purchase logs into structured schemas.

STEP 02

Feature Engineering

Derive age-group cohorts, purchase frequencies, and customer lifecycle segments to enable granular drill-downs.

STEP 03

PostgreSQL Load

Stage trusted records into relational tables and query behavioral slices with clean, documented SQL.

STEP 04

Power BI Delivery

Translate complex SQL outputs into an executive dashboard providing instant visibility into revenue mix.

An end-to-end framework built for continuous ingestion rather than one-off exploratory reporting.

ETL ARCHITECTURE

From messy records to a reusable analytical pipeline



Python cleaning



Feature engineering



PostgreSQL customer table

SQL analysis



Power BI dashboard

37 missing review ratings

Imputed with category medians.

Redundant field removed

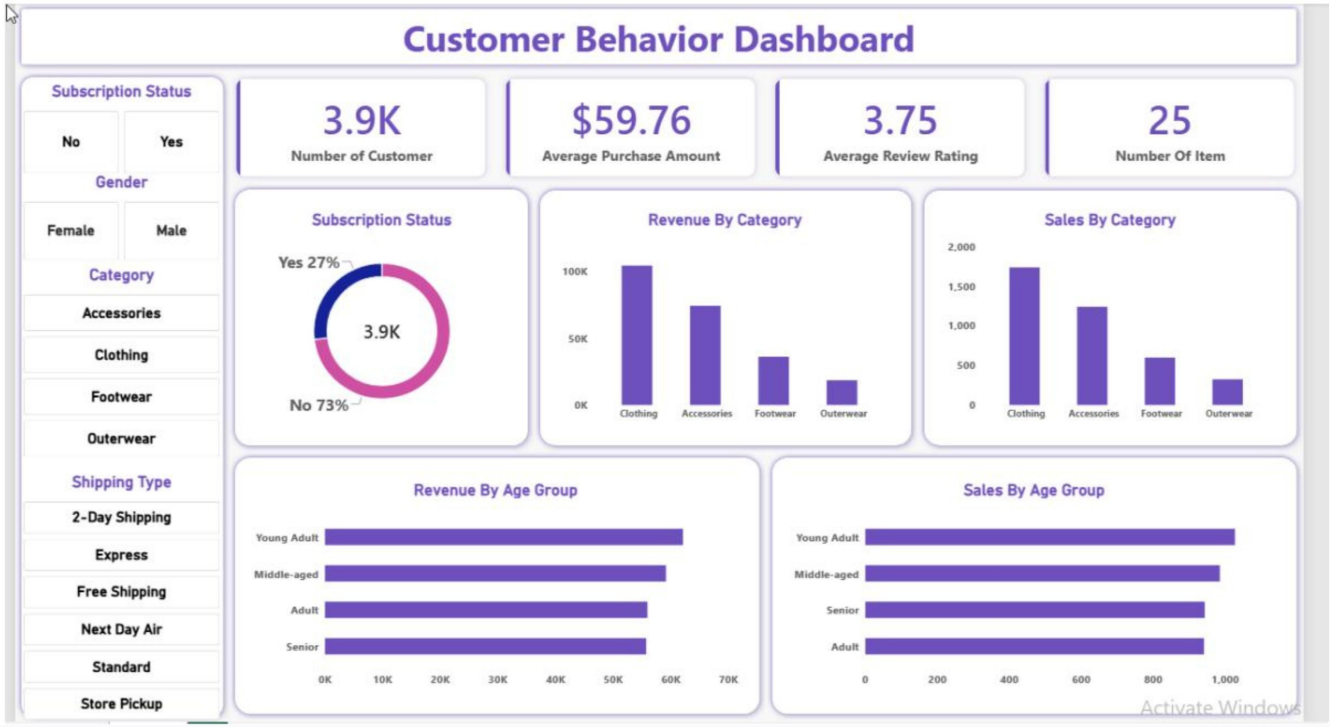
promo_code_used was removed because it mirrored discount_applied.

Segmentation features

Age bands and purchase-frequency days were derived for segmentation.

INTERACTIVE REPORTING

The dashboard makes category and customer patterns visible



Power BI production environment anchored on PostgreSQL curated tables.

Subscription Mix

27% subscriber share is an immediate lever for recurring revenue growth.

Category Anchor

Clothing dominates transaction volume; Accessories follow closely.

Age Dispersion

Revenue spans demographics, not one niche.

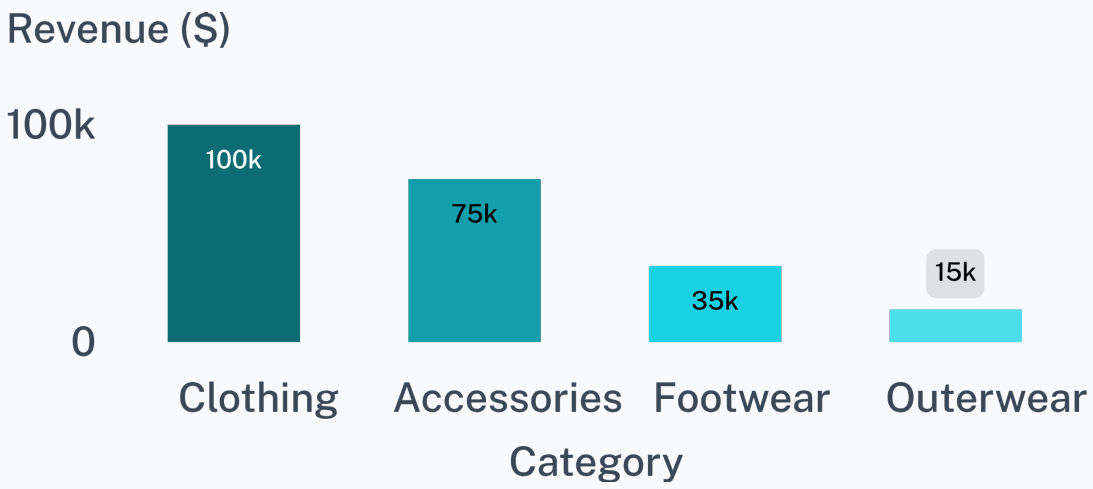
Shipping Slicers

Express and standard show parity in basket sizes across delivery types.

CATEGORY PERFORMANCE

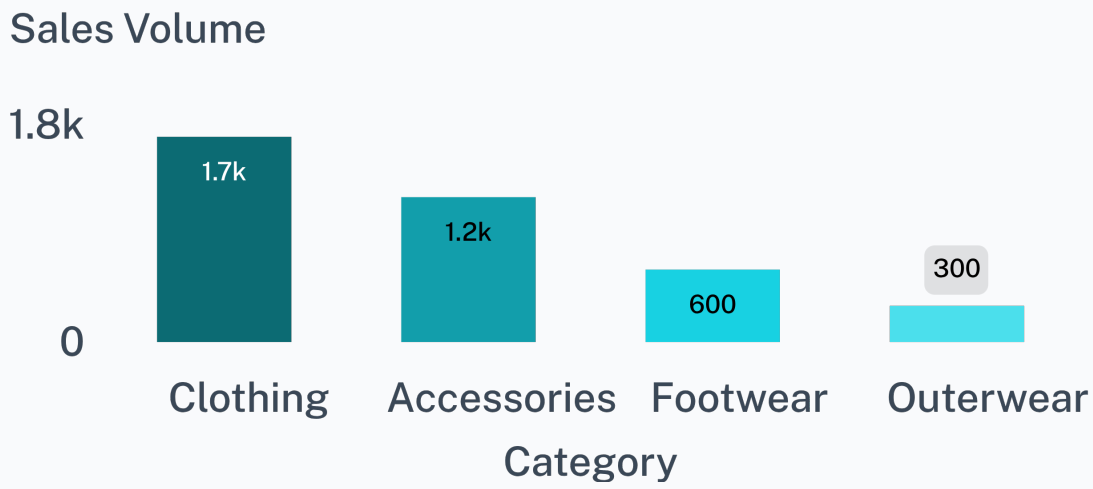
Clothing is the clear revenue and volume anchor

Revenue by Category ($)



Clothing ($100K) and Accessories ($75K) drive 78% of portfolio revenue.

Unit Sales Volume by Category



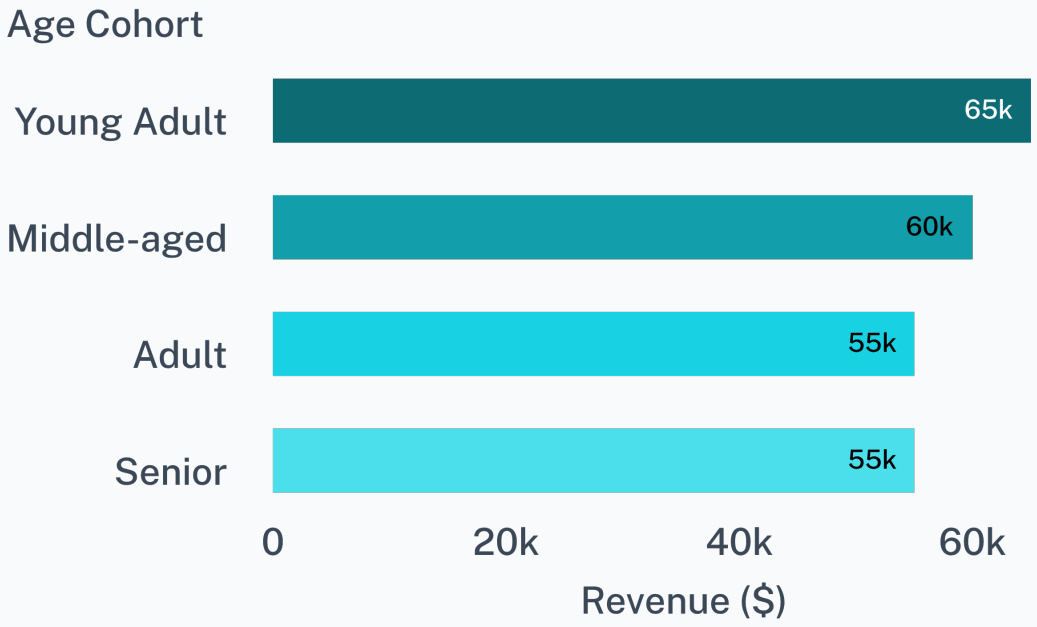
Volume directly mirrors revenue: 1,700 orders originate in Clothing.

Strategic Takeaway: Leverage core Clothing leaders (e.g., Blouses and Sweaters) to cross-sell lower-volume Footwear and Outerwear lines.

DEMOGRAPHIC & LOYALTY BREAKDOWN

Age groups contribute almost evenly, while subscriptions lag

Total Revenue by Age Group ($)



Balanced distribution indicates broad catalog appeal across life stages.

SUBSCRIPTION STATUS

27% Subscribed

73% Non-Subscribed

1,053 Yes vs 2,847 No

Commercial Implication

The business has broad multi-generational appeal, but the loyalty mechanism is not converting enough of that demand.

DEEPER SQL FINDINGS

SQL analysis surfaces the actions behind the dashboard

DIMENSION	ANALYSIS
Gender mix	Male customers represent roughly 68% of transactions, leaving a female-growth opportunity.
Shipping	Standard and express average purchase amounts are virtually indistinguishable, suggesting urgency rather than basket size drives upgrades.
Loyalty	Repeat buyers do not automatically subscribe, so existing incentives are not strong enough.
Merchandising	Top-rated products and category anchor items can power placement and inventory decisions.

STRATEGIC ROADMAP

Three moves can convert insight into growth

MOVE 01

Revamp subscription incentives

Use member-only bundles or threshold-free shipping.

MOVE 02

Cross-sell from category anchors

Use Clothing leaders such as Blouses and Sweaters to drive Accessories and Footwear attach.

MOVE 03

Replace blanket discounts

Use minimum-spend tiers, such as $15 off orders above $75, to lift baskets above the $59.76 baseline.

The next analytical sprint should measure conversion, not just volume

- Track subscription conversion by customer segment, category, and prior-purchase count.
- Test tiered offers against blanket promotions and monitor average order value.
- Build a recurring dashboard view for retention, discount reliance, and cross-category attach.
- Use the SQL layer as a repeatable decision system, not a one-time report.